

## Literature Survey

The rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) has significantly transformed the healthcare industry by enabling early disease prediction, personalized treatment, and preventive healthcare. Several research studies have explored the application of AI in analyzing medical records, symptoms, and lifestyle data to improve healthcare outcomes. The following literature survey summarizes important contributions in this domain.

| S.No | Author(s)<br>& Year                    | Title                                                                           | Methodology                                                                                                             | Limitations                                                                           |
|------|----------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1    | UCI Machine Learning Repository (2020) | Diabetes Prediction Using Machine Learning                                      | Used Logistic Regression, Decision Trees, and Random Forest for diabetes prediction based on patient health parameters. | Focused only on diabetes prediction and lacked personalized health recommendations.   |
| 2    | Esteva et al. (2019)                   | A Guide to Deep Learning in Healthcare                                          | Applied Deep Learning techniques for disease diagnosis using clinical data and medical imaging.                         | Required large datasets and high computational resources.                             |
| 3    | Rajkomar et al. (2018)                 | Scalable and Accurate Deep Learning for Electronic Health Records               | Used Neural Networks to predict diseases from Electronic Health Records (EHR).                                          | Relied heavily on structured hospital datasets and did not consider lifestyle habits. |
| 4    | Topol (2019)                           | High-Performance Medicine: The Convergence of Human and Artificial Intelligence | Discussed AI-assisted healthcare systems for disease diagnosis and personalized treatment recommendations.              | Mainly theoretical with limited implementation details.                               |

|   |                                  |                                       |                                                                                                                  |                                                                       |
|---|----------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 5 | Chawla et al. (2021)             | AI-Based Health Recommendation System | Developed a recommendation system using patient symptoms and medical history for personalized healthcare advice. | Did not provide continuous health monitoring or wellness tracking.    |
| 6 | WHO Digital Health Report (2021) | Digital Health Strategy               | Emphasized AI-driven preventive healthcare, digital monitoring, and patient-centered health management.          | Focused on healthcare policies rather than implementation techniques. |

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## Research Gap

Although existing healthcare systems successfully predict specific diseases, several limitations remain:

- Most systems focus on predicting only one disease.
  - Few applications combine lifestyle habits with symptom analysis.
  - Existing systems generally provide static recommendations instead of personalized wellness plans.
  - Limited support for continuous health monitoring and weekly progress analysis.
  - Most healthcare applications do not include mental wellness, nutrition, hydration, and fitness recommendations together.
  - Root cause analysis explaining why a prediction occurs is rarely included.
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## Proposed Solution

The proposed project, **WellnessIQ**, addresses these limitations by integrating Artificial Intelligence and Machine Learning into a comprehensive healthcare platform.

The system:

- Collects personal, medical, and lifestyle information.
- Analyzes symptoms and health parameters using ML models.
- Predicts potential health risks.
- Generates personalized recommendations for diet, exercise, sleep, hydration, mental wellness, and lifestyle improvement.
- Provides AI-generated root cause analysis to explain possible reasons behind health risks.

- Maintains weekly health records and visualizes progress through interactive dashboards.
  - Sends daily reminders for health check-ins to encourage preventive healthcare.
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## Comparative Analysis

| Parameter                       | Existing Systems | WellnessIQ |
|---------------------------------|------------------|------------|
| Disease Prediction              | ✓                | ✓          |
| Lifestyle Analysis              | Partial          | ✓          |
| Symptom Analysis                | ✓                | ✓          |
| Personalized Diet Plan          | Limited          | ✓          |
| Workout Recommendation          | Limited          | ✓          |
| Mental Wellness Recommendation  | ✗                | ✓          |
| Hair & Skin Care Recommendation | ✗                | ✓          |
| AI Root Cause Analysis          | ✗                | ✓          |
| Weekly Health Tracking          | Limited          | ✓          |
| Health Score Generation         | ✗                | ✓          |
| Interactive Dashboard           | Partial          | ✓          |
| Daily Health Reminder           | ✗                | ✓          |

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## Conclusion

The literature indicates that AI-based healthcare systems have achieved significant success in disease prediction and diagnosis. However, most existing solutions are limited to predicting individual diseases and do not provide a holistic view of a user's health. **WellnessIQ** extends existing research by combining disease prediction, lifestyle analysis, symptom assessment, and AI-powered personalized recommendations into a unified platform. By integrating continuous health monitoring, wellness tracking, and preventive healthcare guidance, the proposed system aims to promote healthier lifestyles and support early intervention for potential health risks.